Data collection

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Definition of data:

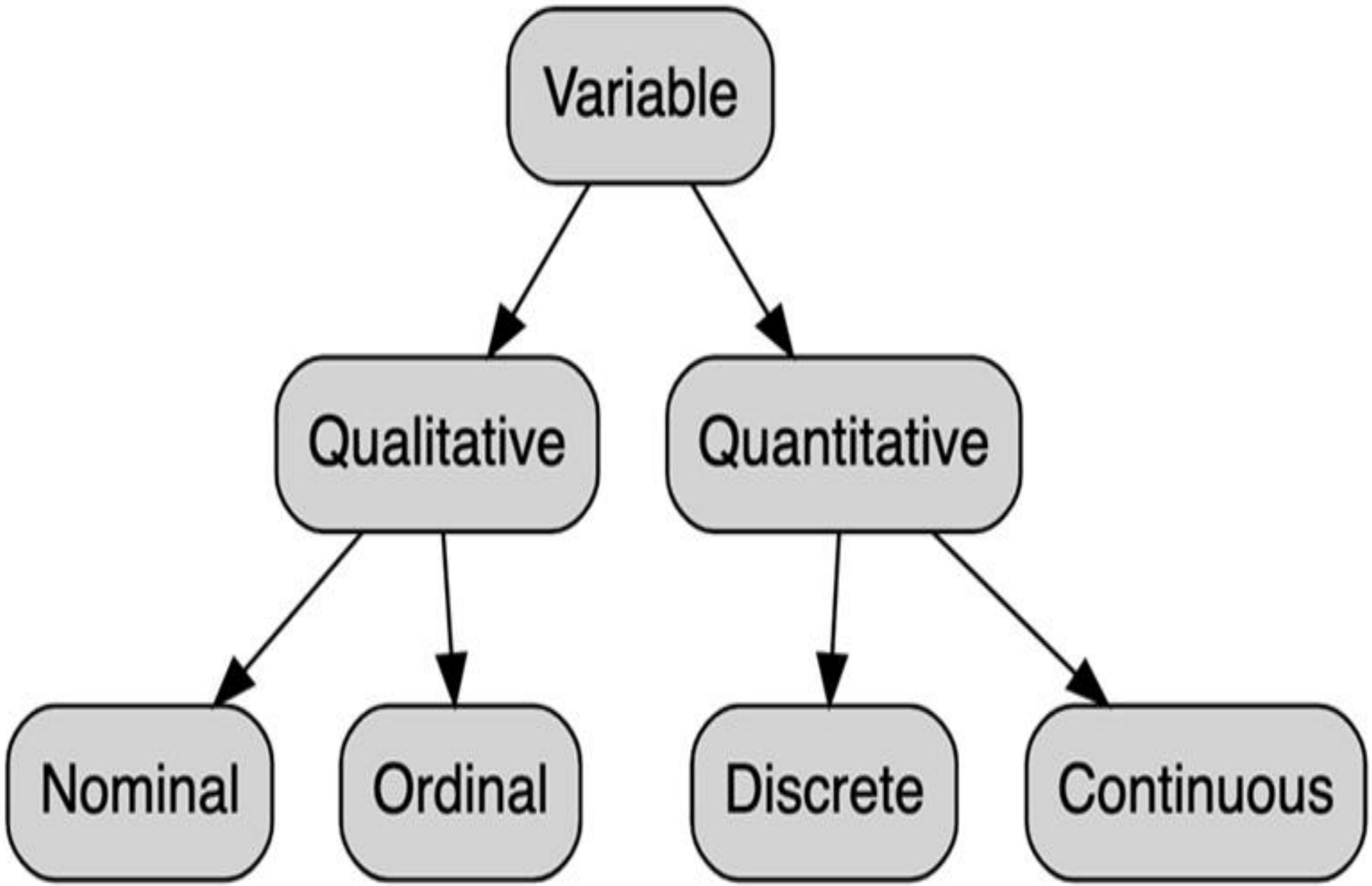
- **It is unprocessed information may be constant or variables data.**

Definition of data collection:

- **Data collection is the process of getting the information (data) that is useful to the study, and which will enable the researcher to answer the research questions and/or test the hypothesis.**

Types of data:

- **There are two types of data.**
- **A) Constant data:** This type is of no statistical importance, such as **normal person has 2 eyes**...etc......
- **B) Variables:** Such as height, weight, blood pressure, pulse rate, blood urea, blood calcium etc.


Variable
Qualitative
Quantitative
Nominal
Ordinal
Discrete
Continuous

There are two types of variables:

- 1 - Quantitative variables.
- 2- Qualitative variables.

1-Quantitative variables:

These are variables which can be expressed in the form of quantities or numbers.

- **They are classified into 2 subtypes:**

Continuous quantitative variables:

- These variables are usually obtained through measurements. i.e. body temperature, blood pressure, height, weights........etc
- **Examples**:
- Height, weight, blood sugar level, blood urea, blood cholesterol, and calcium in blood.

Discrete quantitative variables:

This type of variables is usually obtained **through numeration**. It is always an integer value and never a fractional value **such as pulse rate, number of pregnancies, or deliveries**, or children or attacks of diseases. This group takes the form of qualities or names and can’t be expressed in the form of quantities or numbers.

Examples:

- Number of children per family
- Number of students in a class
- Number of citizens of a country

2) Qualitative variable:

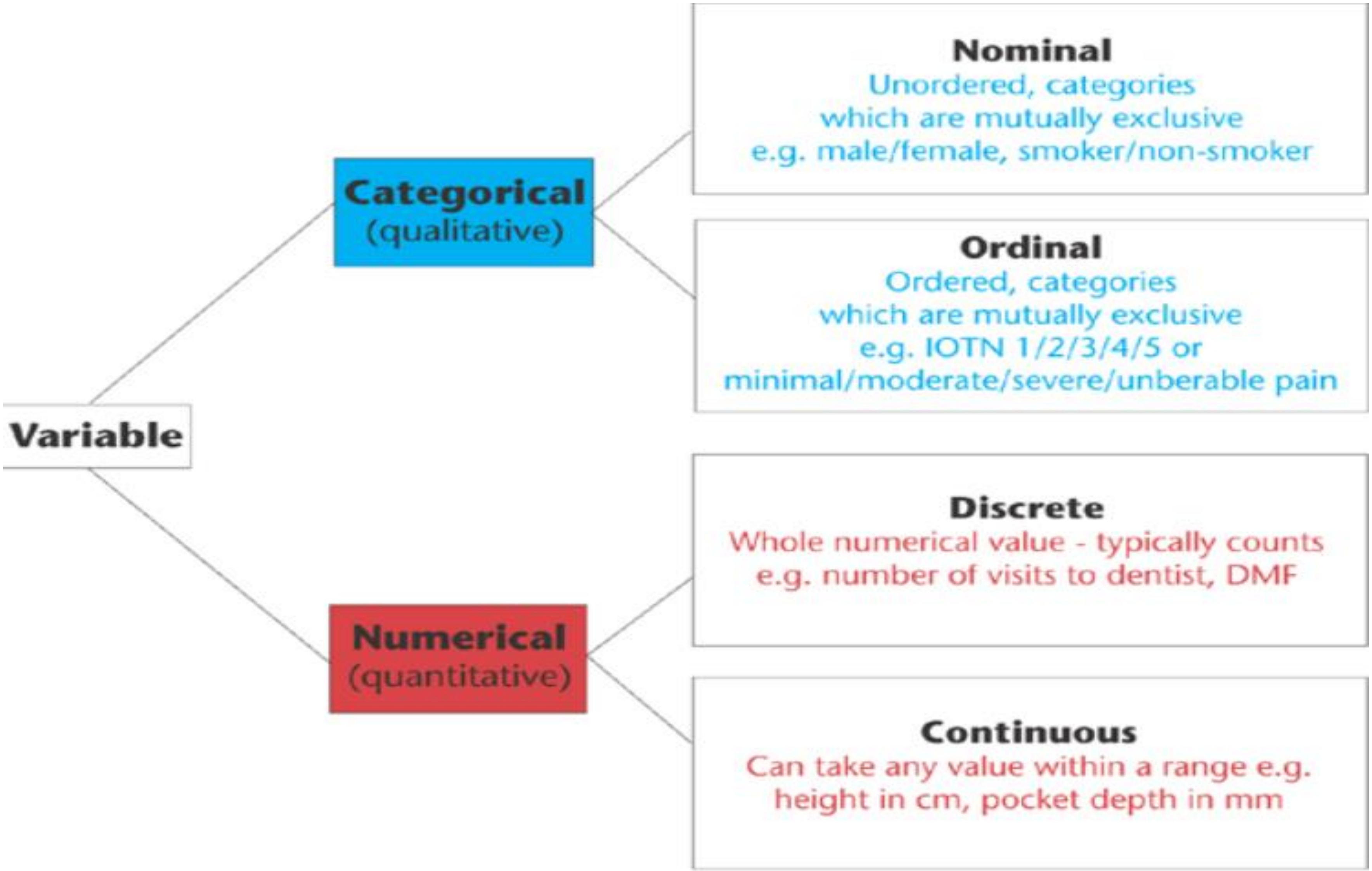
- A qualitative variable, also called **a categorical variable**, is a variable **that isn't numerical**. It describes data that fits into categories. The different values taken on by a categorical variable are often called levels.

Qualitative variable involved two types:

- **Ordinal qualitative variables:**
- An ordinal variable is a categorical variable for which the possible values are ordered. Ordinal variables can be considered "in between" categorical and quantitative variables.
- **Example:**
- **Social levels:** (upper social class, middle social class, lower social class).
- **Levels of education: (**Illiterate, read& write, primary, secondary, university).

B) Nominal qualitative variables

- The other type of qualitative variables in which categories can't be arranged in any definite order is called such as:
- **- Marital status:** single, married, divorced and widow.
- **- Blood groups:** A, B, AB, O.


Variable
Categorical
(qualitative)
Nominal
Unordered, categories
which are mutually exclusive
e.g. male/female, smoker/non-smoker
Ordinal
Ordered, categories
which are mutually exclusive
e.g. IOTN 1/2/3/4/5 or
minimal/moderate/severe/unberable pain
Numerical
(quantitative)
Discrete
Whole numerical value - typically counts
e.g. number of visits to dentist, DMF
Continuous
Can take any value within a range e.g.
height in cm, pocket depth in mm

Types of variables	
Quantitative	Qualitative
Quantitative Variables **It can be measure in the usual sense.** **For example:** **- The heights of adult males.** **- The weights of newborn babies.** **-The ages of patients seen in a dental clinic**	Qualitative Variables Many characteristics are not capable of being measured. Some of them can be ordered or ranked. For example: - Classification of people into socio-economic groups. -Social classes based on income, education, etc

Types of Quantitative Variables	
Discrete	**Continuous**
Is characterized by gaps or interruptions in the values that it can assume. For example: • The number of daily admissions to a governmental hospital. • The number of filled teeth per child in an elementary school.	Can assume any value within a specified relevant interval of values assumed by the variable. For example: • Height, weight, • Skull circumference. • No matter how close together the observed heights of two people, we can find another person whose height falls somewhere in between.

THANK YOU